

Midterm 1

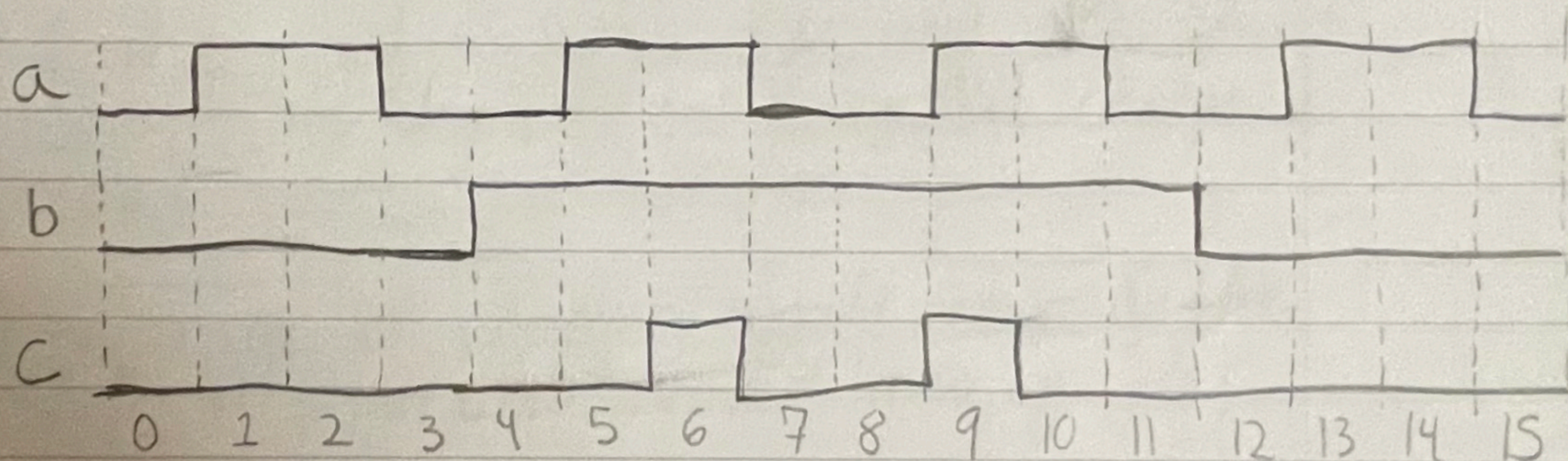
Chapter 1 - introductory concepts

1 Consider figure 1. (a) What is the duty cycle of each D_i signal? (b) Consider the bitstreams of D_i . What does the sequence of numbers represent?

a) D_0 : 50 %
 D_1 : 50 %
 D_2 : 50 %
 D_3 : 50 %

b) D_0 : 0101010101010101/
 D_1 : 0011001100110011/
 D_2 : 0000111100001111/
 D_3 : 0000000011111111/

- 2 (a) Create the timing diagram representing the output of an XOR gate with D_0 and D_1 as inputs.
(b) do the same for D_2 and D_3
(c) Finally, create the Timing diagram representing the output of a 3-input AND gate with the XOR gates from parts (a) and (b), and the EN signal as inputs



3 Suppose D_i represents parallel data with a clock frequency of 4 MHz . (a) what is the total bitrate (bits per second)? (b) what would be the bitrate if the system was serial instead of parallel?

a)

Chapter 2 - Number Systems, Operations, and codes

1 Convert to Binary: a) 1024 b) $0xB3B3$ c) -2048

a) 100000000000 b) 1011 1011 1011 1011 c) 1111 1000 0000 0000

2 Convert to hex: a) 65535 b) 1000 1000 1000 1000

a) FFFF b) 8888

3 Design problem. Consider the angular shaft encoder in fig. 2. The IR emitter/detector detects IR light reflected from white sectors, while the dark sectors absorb the light. The sectors are encoded in binary. For example, starting from inner sectors and reading outward, dark-dark-light is translated to State 001 by the detector. If the shaft rotates 45 degrees, the detector records a state change.

a) if the initial state is 000 and the shaft rotates 360 degrees, how many bit changes occur?

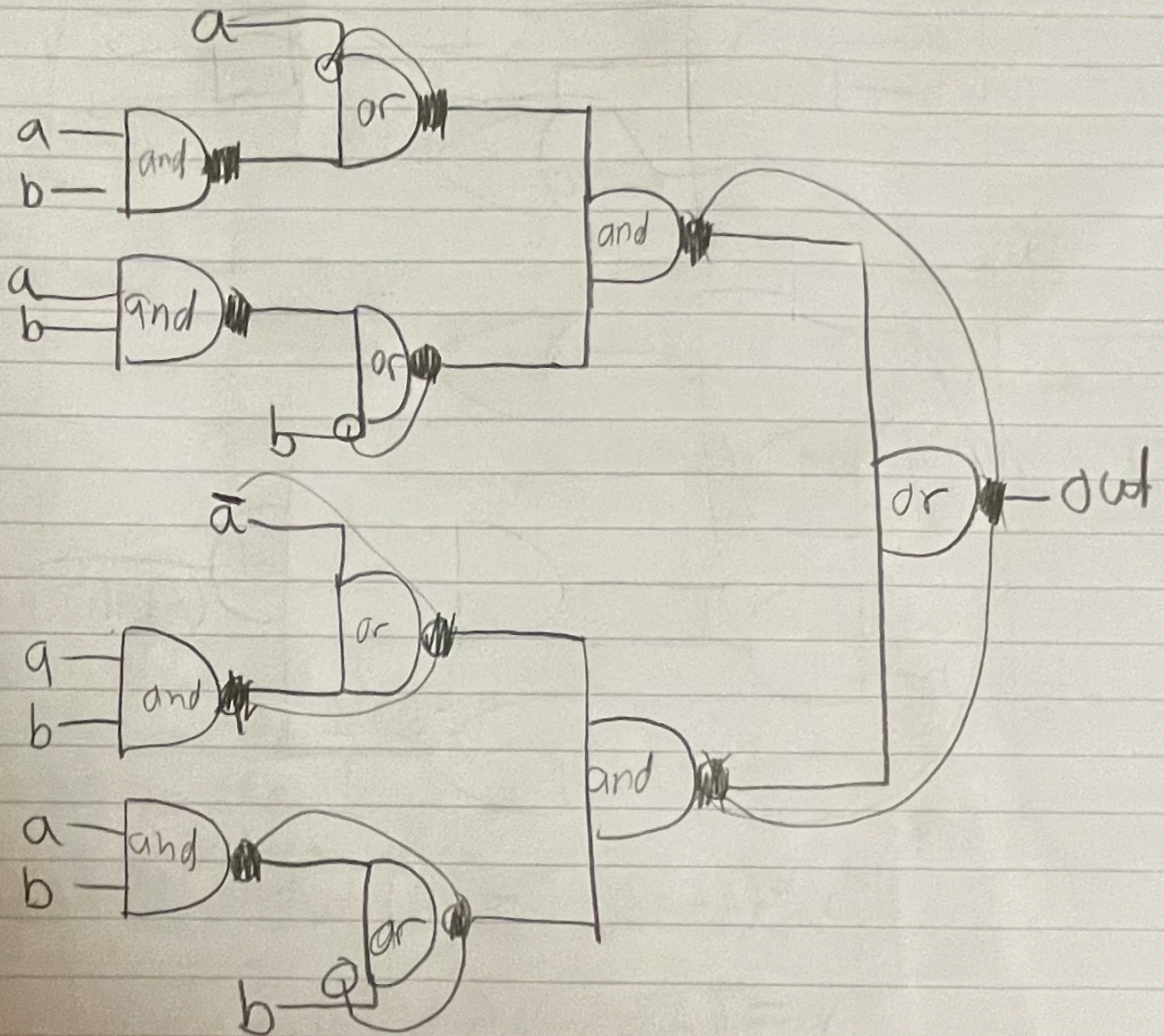
b) How many bit changes would occur if the shaft position was encoded in grey code?

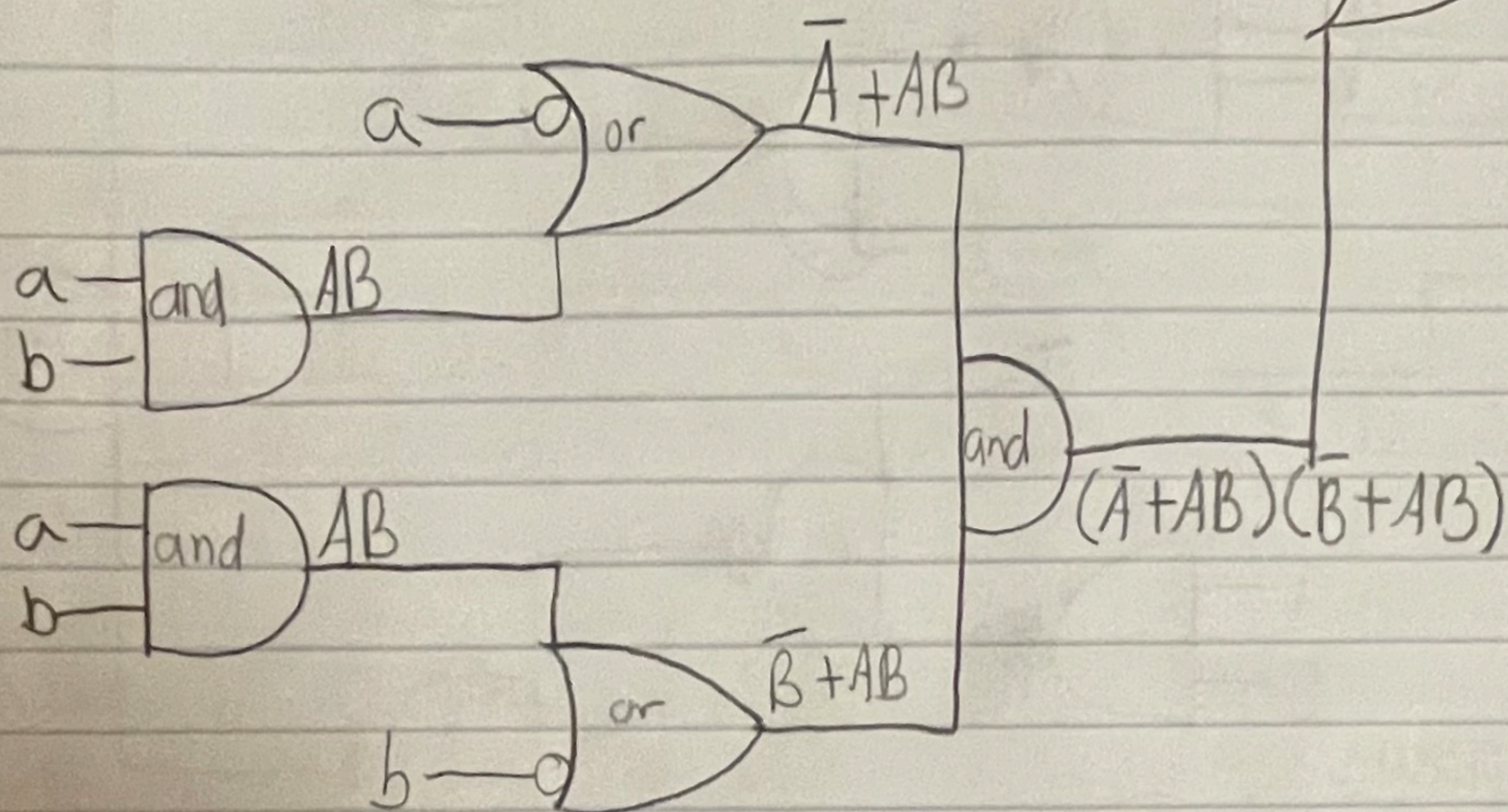
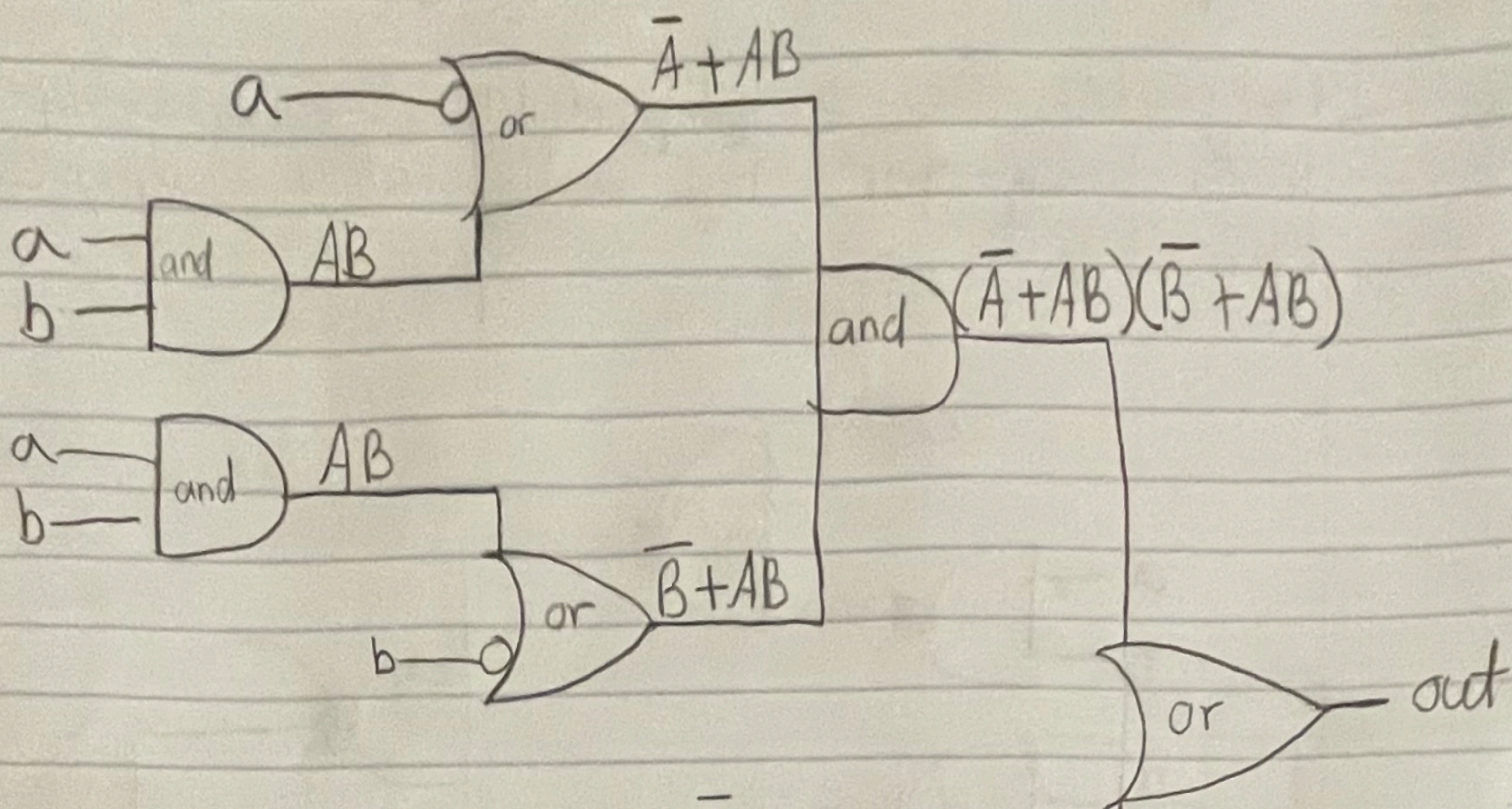
c) Draw the correct face of the shaft encoder, encoded in 3-bit grey code instead of binary¹

d) If 45 degrees is the angular precision with 3-bit grey code, what is the angular precision with 8 bit grey code?

Chapter 3 - Logic Gates

1 Generate the simplified logic expression and truth table for Fig 3, left. What do you call this type of gate?





$$X = (\bar{A} + AB)(\bar{B} + AB) + (\bar{A} + AB)(\bar{B} + AB)$$

$$A + A = A$$

$$X = (\bar{A} + AB)(\bar{B} + AB)$$

$$= (\bar{A} + B)(\bar{B} + A)$$

$$= \bar{A}\bar{B} + \underbrace{\bar{A}A}_{=0} + \underbrace{\bar{B}B}_{=0} + BA$$

$$\bar{A} + AB = \bar{A} + B$$

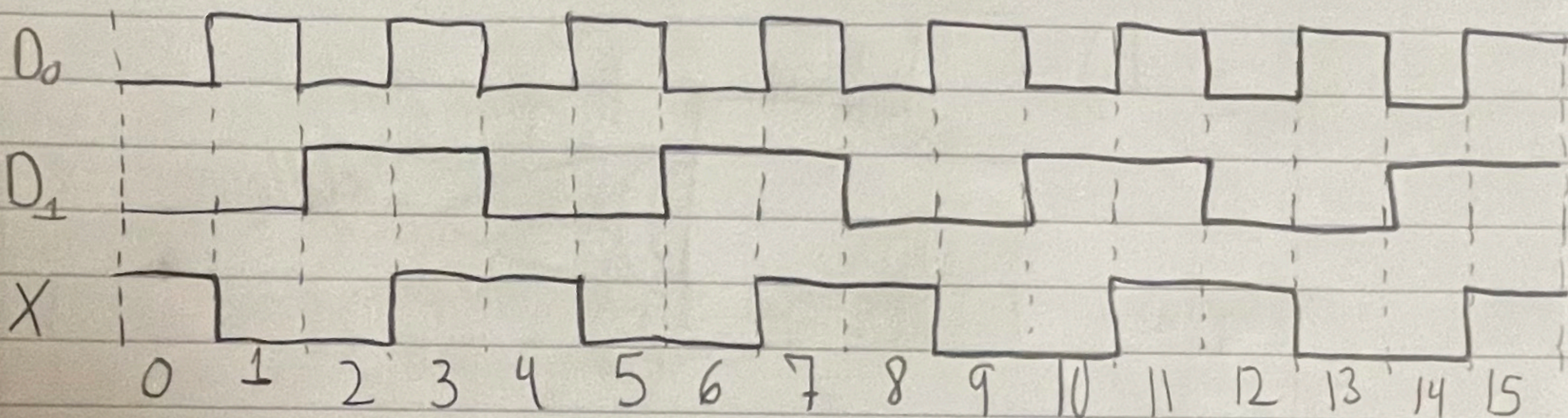
distribute

$$= \bar{A}\bar{B} + BA$$

and or and

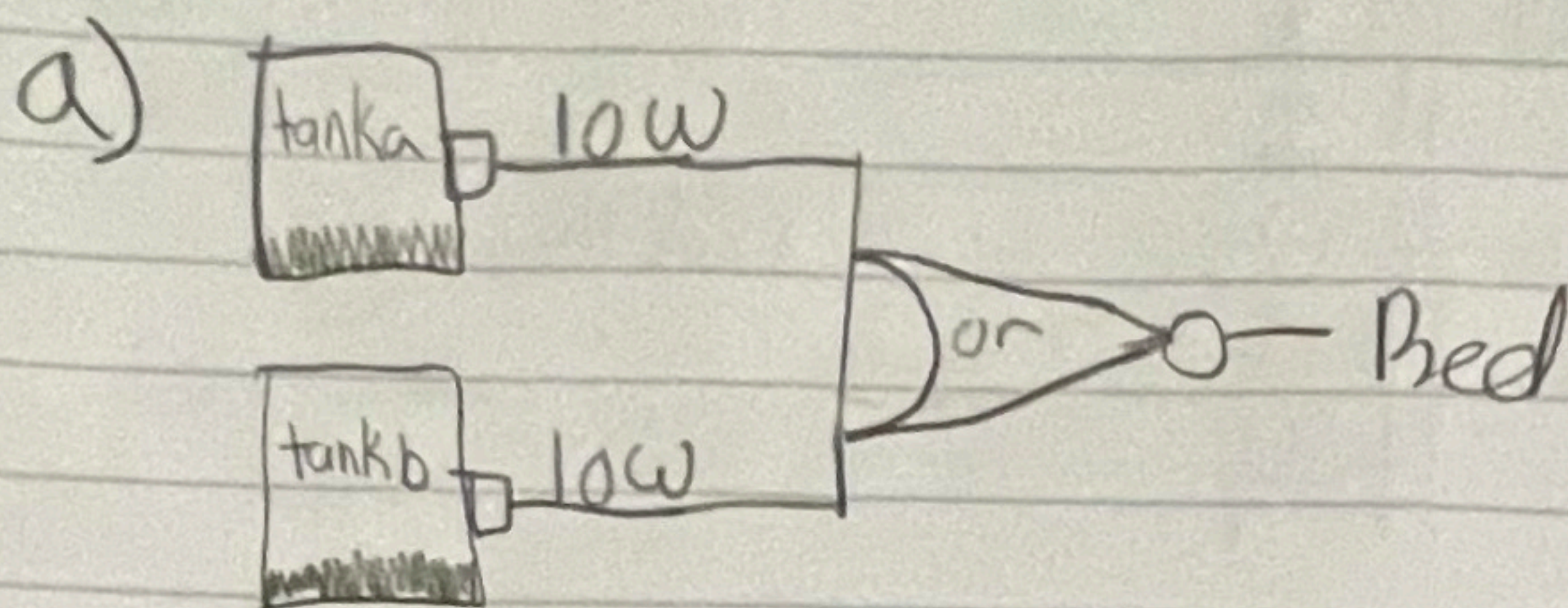
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

XNOR

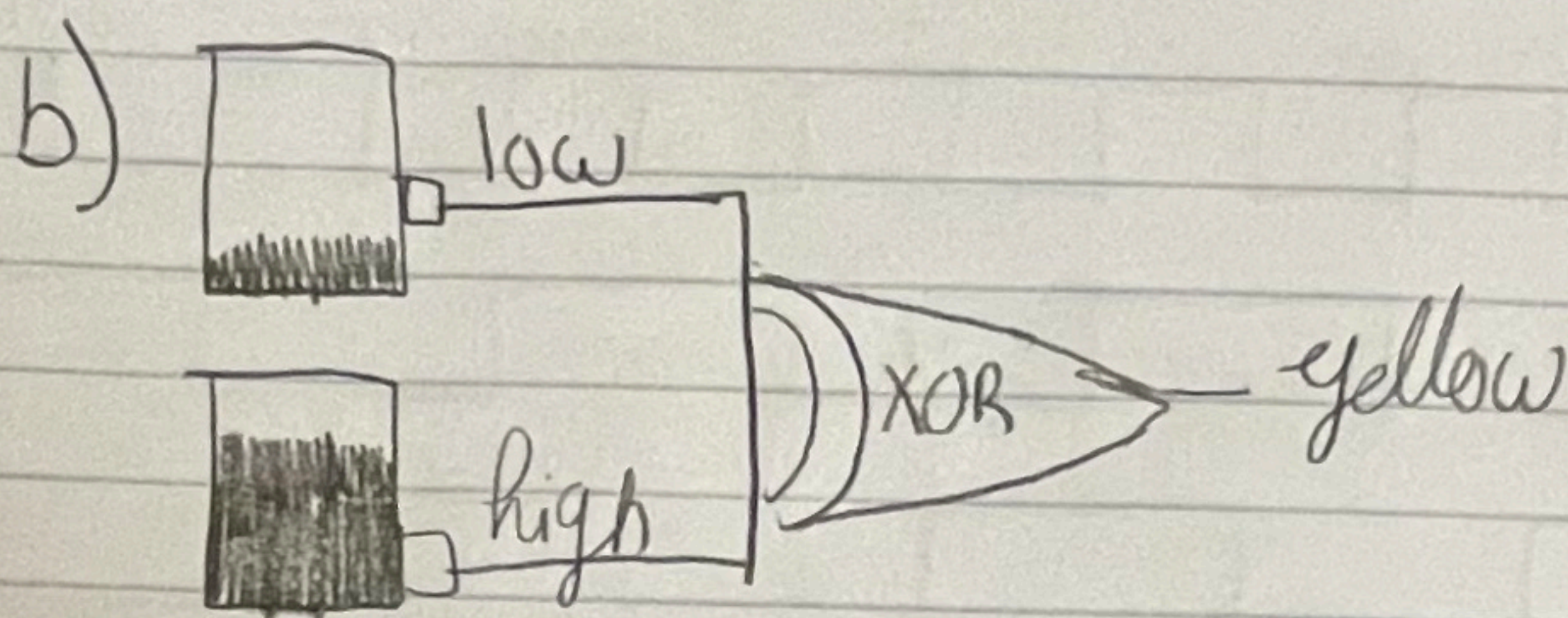


3 Design problem. A liquid tank system is depicted in fig. 3 (right). The sensors are High when the liquid is above the level. a) create and display a red Led system that activates when both tanks are below the level. b) create and display a yellow led system that activates when one tank is below and one tank is above the level. c) Add a third tank with two pipes guiding liquid to tanks A and B. Each pipe should have a valve, add logic that opens the correct valve so as to

fill only the low tank until it is no longer below the level



a	b	X
0	0	1
0	1	0
1	0	0
1	1	0



a	b	X
0	0	0
0	1	1
1	0	1
1	1	0

c)

Chapter 4 - Boolean Algebra and logic simplification

1 Suppose an investment firm manages a portfolio with shares in four stocks, labeled A through D. The companies corresponding to stocks A through D are labeled inactive or active by the firm, based on productivity information. The firm notices that returns (profits) for the portfolio increase under the following conditions:

- All four companies are active
- or
- Companies A and B are inactive, while C and D are active
- or
- Company A is inactive, while B, C, and D are active
- or
- Company B is inactive, while companies A, C, and D are active

a) Develop a S-SOP expression for X, the portfolio's state. If profit is increasing, label the output ON, and label it OFF if profit is decreasing

b) Use a domain-4 Karnaugh map to simplify the S-SOP expression

c) based on your analysis, which stock or stocks should be eliminated from the portfolio?

a) $ABCD + \bar{A}\bar{B}CD + \bar{A}BCD + A\bar{B}CD = 011$

b)

CD \ AB	00	01	10	11
00				1
01				1
10				1
11				1

c) A and B can go away

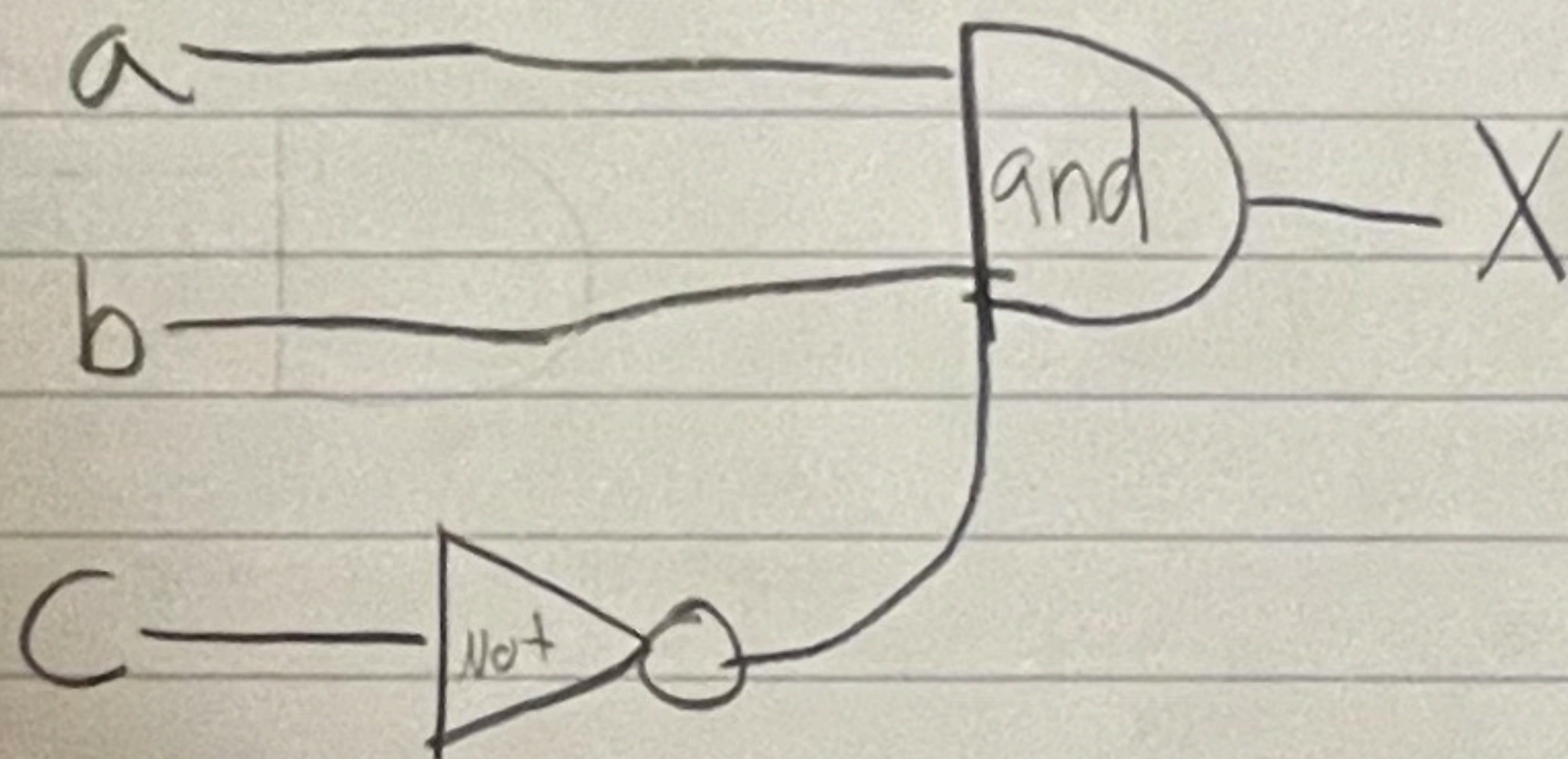
2 A circuit contains three main branches leading to one output. The output is observed to fail under the following conditions below. a) use the domain 3 Karnaugh map to determine the conditions under which it does succeed, and write an S-SOB expression for the circuit. b) draw the circuit using gates.

a)

AB \ C	0	1
00	0	0
01	0	0
10	0	1
11	0	0

$\overline{A}BC$

b)



Chapter 5 - Combinatorial Logic Analysis

1 For the input waveforms shown in fig 4 (left) what logic circuit will generate the output waveform?

